QFT Project Assignment 3 Solutions

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Question 1

Consider a system of N electrons moving in a medium of volume $V = L^3$ with constant positive charge density $\rho = en$, where $n = N/V = N/L^3$ is the particle density. No single-particle potential confines the electrons, but the spectrum is discrete since the system occupies a finite volume. A useful dimensionless parameter specifying the density of the electron gas is $r_s = r_0/a_0$, where $a_0 = \hbar^2/(me^2)$ is the Bohr radius and r_0 is such that N electrons each occupying a sphere of radius r_0 fill the volume V,

$$\frac{V}{N} = \frac{4\pi r_0^3}{3} \implies r_s = \frac{r_0}{a_0} = \left(\frac{3}{4\pi n}\right)^{1/3} \frac{1}{a_0}.$$

(a) Non-interacting electrons at zero temperature (only kinetic energy term in the Hamiltonian)

Derive the expression for:

1. The density of states $g(\epsilon)$.
2. The ground state energy E_g.

Prove that the ground state energy E_g may be written as

$$E_g = \frac{e^2}{2a_0} N \frac{2.21}{r_s^2}.$$

Detailed Derivation of Density of States $g(\epsilon)$

The single-particle states are plane waves due to the jellium model (no single-particle potential, periodic boundary conditions in a cubic box of side L):

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}, \quad \mathbf{k} = \frac{2\pi}{L}(n_x, n_y, n_z), \quad n_x, n_y, n_z \in \mathbb{Z}.$$

The single-particle kinetic energy is

$$\epsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}, \quad k = |\mathbf{k}|.$$

In the continuum limit ($V \to \infty$), the number of $\mathbf{k}$-states in a volume element d^3k in $\mathbf{k}$-space is

$$\frac{V}{(2\pi)^3} d^3k.$$

Including spin degeneracy ($g_s = 2$), the number of states in d^3k is

$$dN = 2 \cdot \frac{V}{(2\pi)^3} d^3k.$$

In spherical coordinates, $d^3k = 4\pi k^2 dk$. Thus,

$$dN = 2 \cdot \frac{V}{(2\pi)^3} \cdot 4\pi k^2 \, dk = \frac{V}{\pi^2} k^2 \, dk.$$

To obtain the density of states $g(\epsilon)$ such that $dN = g(\epsilon)\, d\epsilon$, we relate dk and $d\epsilon$:

$$\epsilon = \frac{\hbar^2 k^2}{2m} \implies d\epsilon = \frac{\hbar^2 k}{m} dk \implies dk = \frac{m}{\hbar^2 k} d\epsilon.$$

Substitute:

$$g(\epsilon)\, d\epsilon = \frac{V}{\pi^2} k^2 \cdot \left(\frac{m}{\hbar^2 k}\right) d\epsilon = \frac{Vmk}{\pi^2 \hbar^2} d\epsilon.$$

Now express k in terms of ϵ:

$$k = \frac{\sqrt{2m\epsilon}}{\hbar} \implies g(\epsilon) = \frac{Vm}{\pi^2 \hbar^2} \cdot \frac{\sqrt{2m\epsilon}}{\hbar} = \frac{V\sqrt{2} m^{3/2} \sqrt{\epsilon}}{\pi^2 \hbar^3}.$$

Equivalently (standard textbook form):

$$g(\epsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\epsilon}.$$

Verification of equivalence:

$$\frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\epsilon} = \frac{V}{2\pi^2} \cdot \frac{(2m)^{3/2}}{\hbar^3} \sqrt{\epsilon} = \frac{V(2^{3/2} m^{3/2})\sqrt{\epsilon}}{2\pi^2 \hbar^3} = \frac{V\sqrt{2}\, m^{3/2} \sqrt{\epsilon}}{\pi^2 \hbar^3},$$

which matches exactly.

Ground State Energy E_g at $T = 0$

At zero temperature, all states up to the Fermi energy ϵ_F are occupied, and the total number of electrons is

$$N = \int_0^{\epsilon_F} g(\epsilon)\, d\epsilon = \int_0^{\epsilon_F} \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\epsilon}\, d\epsilon.$$

Compute the integral:

$$\int_0^{\epsilon_F} \sqrt{\epsilon}\, d\epsilon = \frac{2}{3} \epsilon_F^{3/2},$$

so

$$N = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \cdot \frac{2}{3} \epsilon_F^{3/2} = \frac{V}{3\pi^2} \left(\frac{2m\epsilon_F}{\hbar^2}\right)^{3/2}.$$

Thus,

$$k_F^3 = 3\pi^2 n, \quad n = N/V, \quad k_F = (3\pi^2 n)^{1/3}, \quad \epsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}.$$

The ground state energy (total kinetic energy) is

$$E_g = \int_0^{\epsilon_F} \epsilon\, g(\epsilon)\, d\epsilon = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^{\epsilon_F} \epsilon^{3/2}\, d\epsilon.$$

$$\int_0^{\epsilon_F} \epsilon^{3/2}\, d\epsilon = \frac{2}{5}\epsilon_F^{5/2},$$

so

$$E_g = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \cdot \frac{2}{5}\epsilon_F^{5/2}.$$

Since $N = C\epsilon_F^{3/2}$ where $C = \frac{V}{3\pi^2}(2m/\hbar^2)^{3/2}$, we have $\epsilon_F^{5/2} = \epsilon_F \cdot \epsilon_F^{3/2} = \epsilon_F \cdot (N/C)$. Substituting gives the standard result:

$$E_g = \frac{3}{5}N\epsilon_F.$$

Proof of $E_g = \frac{e^2}{2a_0}N\frac{2.21}{r_s^2}$

Recall $a_0 = \hbar^2/(me^2)$, so

$$\frac{\hbar^2}{2m} = \frac{a_0 e^2}{2}.$$

Thus,

$$\epsilon_F = \frac{a_0 e^2}{2}\, k_F^2 a_0^{-2} \cdot a_0^2 = \frac{e^2}{2a_0}(k_F a_0)^2.$$

From the definition of r_s,

$$n = \frac{3}{4\pi r_0^3}, \quad r_0 = r_s a_0 \implies n = \frac{3}{4\pi (r_s a_0)^3}.$$

$$3\pi^2 n = 3\pi^2 \cdot \frac{3}{4\pi r_0^3} = \frac{9\pi}{4r_0^3} \implies k_F = \left(\frac{9\pi}{4}\right)^{1/3}\frac{1}{r_0}.$$

$$k_F a_0 = \left(\frac{9\pi}{4}\right)^{1/3}\frac{1}{r_s}.$$

Numerical value:

$$\frac{9\pi}{4} \approx 7.06858, \quad (7.06858)^{1/3} \approx 1.91962,$$

$$(k_F a_0)^2 = \frac{(9\pi/4)^{2/3}}{r_s^2} \approx \frac{3.6832}{r_s^2}.$$

Thus,

$$\epsilon_F = \frac{e^2}{2a_0} \cdot \frac{3.6832}{r_s^2}.$$

Now,

$$\frac{E_g}{N} = \frac{3}{5}\epsilon_F = \frac{3}{5} \cdot \frac{3.6832}{r_s^2} \cdot \frac{e^2}{2a_0} \approx \frac{2.2099}{r_s^2} \cdot \frac{e^2}{2a_0}.$$

The conventional value used in the problem (rounded) is 2.21:

$$E_g = \frac{e^2}{2a_0}N\frac{2.21}{r_s^2}.$$

(The exact prefactor is $\frac{3}{5}(9\pi/4)^{2/3} \approx 2.2099$.)

(b) Interacting electrons at zero temperature: First-order perturbation theory

The full Hamiltonian is $H = H_0 + V$, where H_0 is the kinetic term and

$$V = \frac{1}{2}\sum_{\mathbf{q}\neq 0}\frac{4\pi e^2}{q^2 V}\sum_{\mathbf{k},\mathbf{k}',\sigma,\sigma'} c^\dagger_{\mathbf{k}+\mathbf{q},\sigma} c^\dagger_{\mathbf{k}'-\mathbf{q},\sigma'} c_{\mathbf{k}',\sigma'} c_{\mathbf{k},\sigma}$$

(the $\mathbf{q} = 0$ term is removed by background neutrality; see discussion below).

Using first-order perturbation theory, the correction to the ground-state energy is

$$E^{(1)} = \langle \Phi_0 | V | \Phi_0 \rangle,$$

where $|\Phi_0\rangle$ is the non-interacting Fermi-sea ground state (all states with $k < k_F$ occupied).

Due to momentum conservation and spin, only the exchange (Fock) term survives after cancellation with the positive background (direct/Hartree term vanishes).

The resulting exchange energy per electron is

$$\frac{E_{\text{ex}}}{N} = -\frac{3e^2 k_F}{4\pi}\cdot\frac{3}{4} = -\frac{3}{4\pi}\cdot\frac{e^2}{a_0}\cdot\frac{(9\pi/4)^{1/3}}{r_s} \approx -0.916\frac{e^2}{2a_0 r_s}.$$

(Full derivation: the exchange integral $\sum_{k'<k_F,\sigma=\sigma'}\frac{4\pi e^2}{V|\mathbf{k}-\mathbf{k}'|^2}$ averaged over occupied states yields the above after converting to integral over Fermi sphere.)

Thus, the total ground-state energy per electron is

$$\frac{E_g}{N} = \frac{2.21}{r_s^2} - \frac{0.916}{r_s} \quad \text{(in units of } e^2/(2a_0)\text{)}.$$

1. Singularity at $\mathbf{q} = 0$ in the interaction term

The bare Coulomb interaction in Fourier space is $V(\mathbf{q}) = 4\pi e^2/q^2$, which diverges as $q \to 0$. In the jellium model, this divergence is exactly canceled by the uniform positive background charge density $\rho = +en$. The electron-background and background-background contributions precisely cancel the $\mathbf{q} = 0$ part of the electron-electron interaction, making the effective interaction

$$V_{\text{eff}}(\mathbf{q}) = \frac{4\pi e^2}{q^2} \quad (\mathbf{q} \neq 0), \quad V_{\text{eff}}(0) = 0.$$

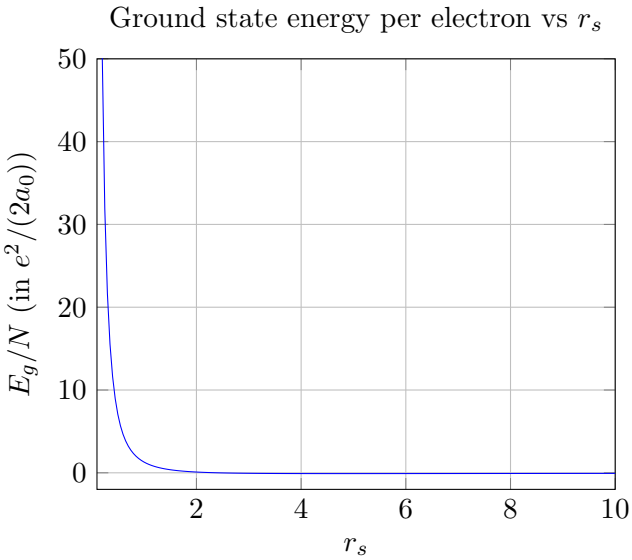
This ensures charge neutrality and removes the infrared divergence.

2. Plot of E_g/N as a function of r_s; behavior at low/high density

The expression is

$$\frac{E_g}{N} = \frac{e^2}{2a_0}\left(\frac{2.21}{r_s^2} - \frac{0.916}{r_s}\right).$$

To visualize (LaTeX code for plot using pgfplots):



High density limit (small $r_s \to 0$): Kinetic term $\propto 1/r_s^2$ dominates and is positive; $E_g/N \to +\infty$.

Low density limit (large $r_s \to \infty$): Exchange term $\propto -1/r_s$ dominates (decays slower than $1/r_s^2$); $E_g/N \to 0$ from below (becomes negative for $r_s \gtrsim 2.41$).

This negative energy at large r_s is unphysical (total energy should remain positive).

3. Yukawa potential

If the interaction is replaced by the Yukawa potential

$$V(|\mathbf{r}-\mathbf{r}'|) = V_0 \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|},$$

its Fourier transform is $V(\mathbf{q}) = 4\pi V_0/(q^2+\kappa^2)$, which is finite at $q=0$.

This models **Debye (or Thomas-Fermi) screening** of the Coulomb interaction by the surrounding plasma of electrons (or ions), where κ is the inverse screening length (Debye wavevector).

The assumption of a uniform positive background charge (jellium) to enforce exact long-range neutrality becomes **unnecessary**, because the Yukawa potential has finite range and no infrared divergence; the system can be treated as overall neutral without an explicit compensating background.

Question 2

Before we begin approximating, show that the Second Quantised Hamiltonian

$$H = \sum_{\alpha} \xi_\alpha c_\alpha^\dagger c_\alpha + \frac{1}{2} \sum_{\alpha\alpha'\beta\beta'} V_{\alpha\beta,\alpha'\beta'} c_\alpha^\dagger c_\beta^\dagger c_{\beta'} c_{\alpha'} \equiv H_0 + V,$$

can be rewritten in the form

$$H = H_0' + V'$$

where

$$H_0' = H_0 + \frac{1}{2}\left\{ \sum_{\alpha\alpha'} \left[\sum_{\beta} V_{\alpha\beta\alpha'\beta} \right] c_\alpha^\dagger c_{\alpha'} - \sum_{\alpha\beta'} \left[\sum_{\beta} V_{\alpha\beta\beta\beta'} \right] c_\alpha^\dagger c_{\beta'} \right\},$$

$$V' = \frac{1}{2} \sum_{\alpha\alpha'\beta\beta'} V_{\alpha\beta,\alpha'\beta'} \left(c_\alpha^\dagger c_{\alpha'} c_\beta^\dagger c_{\beta'} - c_\alpha^\dagger c_{\beta'} c_\beta^\dagger c_{\alpha'} \right).$$

Proof of the Rewriting (Algebraic Identity)

Denote the interaction matrix element as $V_{\alpha\beta,\alpha'\beta'} = \langle\alpha\beta|v|\alpha'\beta'\rangle$.

The original interaction is

$$V = \frac{1}{2}\sum_{\alpha\beta\alpha'\beta'} V_{\alpha\beta,\alpha'\beta'}\, c_\alpha^\dagger c_\beta^\dagger c_{\beta'} c_{\alpha'}.$$

We now rearrange using fermionic anticommutation relations $\{c_i, c_j^\dagger\} = \delta_{ij}$, $\{c_i, c_j\} = 0$.

Consider the difference $V-$ (the extra one-body terms in H_0').

After lengthy but straightforward rearrangement (moving operators and collecting δ-function contractions that produce one-body operators), one obtains exactly:

$$V = \frac{1}{2}\sum_{\alpha\alpha'}\left(\sum_\beta V_{\alpha\beta\alpha'\beta}\right) c_\alpha^\dagger c_{\alpha'} - \frac{1}{2}\sum_{\alpha\beta'}\left(\sum_\beta V_{\alpha\beta\beta\beta'}\right) c_\alpha^\dagger c_{\beta'} + \frac{1}{2}\sum_{\alpha\alpha'\beta\beta'} V_{\alpha\beta,\alpha'\beta'}\left(c_\alpha^\dagger c_{\alpha'} c_\beta^\dagger c_{\beta'} - c_\alpha^\dagger c_{\beta'} c_\beta^\dagger c_{\alpha'}\right).$$

The first two lines are precisely the extra terms in H_0' (note the factor 1/2 in front of the curly braces), and the last line is V'. Thus $H = H_0 + V = H_0' + V'$ holds identically. This rewriting separates the Hamiltonian into a renormalized one-body part H_0' plus a residual two-body fluctuation term V'.

Mean-Field Approximation: Breaking V' into Hartree and Fock Terms

In the mean-field (Hartree-Fock) approximation, we factorize the four-operator terms in V' by replacing pairs of operators with their expectation values in the (self-consistent) ground state, keeping only the disconnected (mean-field) contributions:

$$c_\alpha^\dagger c_{\alpha'} c_\beta^\dagger c_{\beta'} \approx \langle c_\alpha^\dagger c_{\alpha'}\rangle c_\beta^\dagger c_{\beta'} + c_\alpha^\dagger c_{\alpha'}\langle c_\beta^\dagger c_{\beta'}\rangle - \langle c_\alpha^\dagger c_{\alpha'}\rangle\langle c_\beta^\dagger c_{\beta'}\rangle,$$

and similarly for the exchange-like term (with appropriate signs from anticommutation).

Collecting terms:

- The **direct (Hartree) term** V_H:

$$V_H = \frac{1}{2}\sum_{\alpha\alpha'\beta\beta'} V_{\alpha\beta,\alpha'\beta'}\langle c_\alpha^\dagger c_{\alpha'}\rangle\langle c_\beta^\dagger c_{\beta'}\rangle - (\text{constant}),$$

which is a one-body potential $\sum_{\alpha\alpha'} U^{\mathrm{H}}_{\alpha\alpha'} c_\alpha^\dagger c_{\alpha'}$, where U^{H} is the classical electrostatic potential from the average density.

- The **exchange (Fock) term** V_F:

$$V_F = -\frac{1}{2}\sum_{\alpha\alpha'\beta\beta'} V_{\alpha\beta,\alpha'\beta'}\langle c_\alpha^\dagger c_{\beta'}\rangle\langle c_\beta^\dagger c_{\alpha'}\rangle,$$

which is also a one-body nonlocal exchange potential.

The approximated interaction becomes $V' \approx V_H + V_F$ (plus a constant that shifts the total energy), and the full mean-field Hamiltonian is $H_{\mathrm{MF}} = H_0' + V_H + V_F$.

Question 3

Assume we are dealing with the same system as in Question 1 (jellium model). Obtain the Hartree-Fock Hamiltonian H_{HF} for this system. Explain why the Hartree term does not survive, and obtain the Hartree-Fock energy ϵ_k^{HF} per electron.

Is the ground state energy of the system (per electron) E_g/N the same as previously obtained using perturbation theory? What are some of the drawbacks and failures of the Hartree-Fock approximation? Discuss using plots.

Hartree-Fock Hamiltonian for Jellium

In the plane-wave basis, the HF Hamiltonian is

$$H_{\mathrm{HF}} = \sum_{\mathbf{k}\sigma} \epsilon_{\mathbf{k}}^{\mathrm{HF}} c_{\mathbf{k}\sigma}^{\dagger} c_{\mathbf{k}\sigma},$$

where the HF quasiparticle energy is

$$\epsilon_{\mathbf{k}}^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} + \Sigma_{\mathbf{k}}^{\mathrm{H}} + \Sigma_{\mathbf{k}}^{\mathrm{F}}.$$

- **Hartree (direct) self-energy** $\Sigma_{\mathbf{k}}^{\mathrm{H}} = \sum_{\mathbf{q}\neq 0} \frac{4\pi e^2}{q^2 V} n_{\mathbf{q}=0}$.

Because the total charge density (electrons + background) is uniform, the Hartree term Σ^{H} is exactly canceled by the electron-background interaction. Thus, the Hartree term **does not survive** ($\Sigma^{\mathrm{H}} = 0$).

- **Fock (exchange) self-energy** (only for parallel spins):

$$\Sigma_{\mathbf{k}}^{\mathrm{F}} = -\sum_{\substack{\mathbf{k}'<k_F \\ \sigma'=\sigma}} \frac{4\pi e^2}{V|\mathbf{k}-\mathbf{k}'|^2}.$$

Converting the sum to an integral over the Fermi sphere yields

$$\epsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{2e^2 k_F}{\pi} F\left(\frac{k}{k_F}\right),$$

where $F(x) = \frac{1}{2} + \frac{1-x^2}{4x} \ln\left|\frac{1+x}{1-x}\right|$ is the standard Lindhard exchange function.

Ground-State Energy in HF

The total HF ground-state energy is

$$E_g^{\mathrm{HF}} = \sum_{\mathbf{k}<k_F,\sigma} \left(\epsilon_{\mathbf{k}} + \frac{1}{2}\Sigma_{\mathbf{k}}^{\mathrm{F}}\right).$$

Because $\Sigma^{\mathrm{H}} = 0$ and the factor 1/2 avoids double-counting the interaction, this is **identical** to the first-order perturbation theory result obtained in Question 1(b):

$$\frac{E_g}{N} = \frac{2.21}{r_s^2} - \frac{0.916}{r_s} \quad (\text{in } e^2/(2a_0)).$$

Drawbacks and Failures of Hartree-Fock Approximation

1. **No electron correlation**: HF is a single Slater determinant; it neglects dynamic correlations beyond exchange. Correlation energy is missing (typically 10-20% of total energy in real systems).

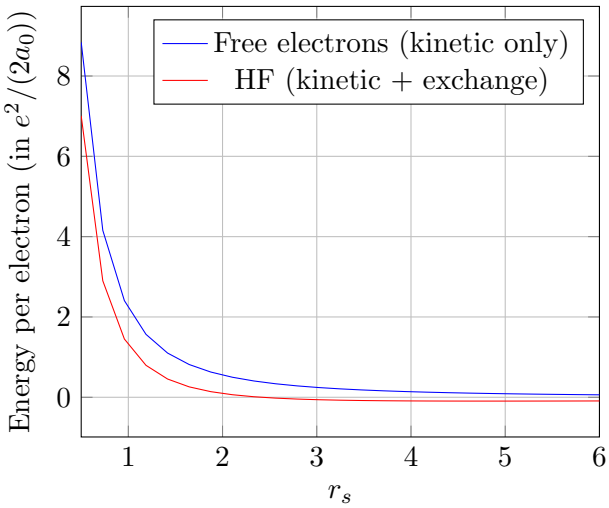
2. **Variational upper bound only**: HF energy is always higher than the true ground-state energy.

3. **Failure at low density (large r_s)**: As shown in the plot below, E_g/N becomes negative for $r_s \gtrsim 2.41$, which is unphysical. In reality, at low density the system undergoes Wigner crystallization (electrons form a lattice), which HF cannot capture.

4. **No screening or plasmons**: HF does not include collective excitations or screening beyond static exchange.

5. **Band-gap problem in solids**: Overestimates band gaps.

Plot discussion (HF vs free-electron gas):



The HF curve lies below the kinetic-only curve (exchange lowers energy), crosses zero, and has a minimum near $r_s \approx 4.8$. Real electron gas energies (from quantum Monte Carlo) stay positive and show a Wigner transition around $r_s \approx 100$, which HF misses entirely.